SIMPLE TWISTED GROUP ALGEBRAS OVER ARBITRARY FIELDS

Abstract. A proof is proposed that a finite group admitting a simple twisted group algebra over an arbitrary field is soluble. The argument establishes a stronger assertion about transitive actions on irreducible characters above a faithful central character, allowing group automorphisms and cyclotomic Galois automorphisms to act together. Ordinary characters and 2-Brauer characters are treated by one minimal-counterexample argument. An affine classification excludes the Brauer case and forces the ordinary case to have radical quotient a direct product of copies of $\mathrm{GL}_3(2)$. Canonical lifts give Galois stability in this class. Restriction to odd-order elements, followed by a central quotient, then produces a smaller counterexample in characteristic two. Galois descent and a Witt-vector lift connect the character theorem to twisted group algebras in every characteristic.

Research preprint. The proposed proof requires independent verification.

1. Introduction

Let G be a finite group and let K be a field. A normalized 2-cocycle is a function $\alpha : G \times G \longrightarrow K^\times$ satisfying

$$\alpha(1, g) = \alpha(g, 1) = 1, \qquad \alpha(g, h)\alpha(gh, x) = \alpha(h, x)\alpha(g, hx).$$

The twisted group algebra $K^\alpha G$ has a K-basis $\{u_g : g \in G\}$ and multiplication $u_g u_h = \alpha(g, h)u_{gh}$. Simplicity means that its only two-sided ideals are zero and the whole algebra; the center need not equal K.

Passman's question, recorded as Problem 18.78 in the *Kourovka Notebook* [7], asks whether simplicity of $K^\alpha G$ forces G to be soluble. The distinction between simple and central simple algebras is essential. In characteristic zero, the central simple case follows from the theorem on groups of central type, whereas an arbitrary simple algebra can split into several simple components after extending the field [9].

The purpose of this article is to give a proposed proof of the following statement.

Theorem 1.1. *Let G be a finite group, let K be a field, and let $\alpha : G \times G \longrightarrow K^\times$ be a normalized 2-cocycle. If $K^\alpha G$ is simple, then G is soluble.*

The character-theoretic statement underlying the proof permits group automorphisms and Galois automorphisms to act together. Write $\mathbb{Q}^{\mathrm{ab}}$ for the union of the cyclotomic fields in $\mathbb{C}$, and put $\Omega = \mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$. For $a \in \mathrm{Aut}(D)$, $\sigma \in \Omega$, and an ordinary character χ, the paired action is

$$\chi^{(a,\sigma)}(g) = \sigma(\chi(a^{-1}(g))).$$

The two operations commute. If $C \leq Z(D)$ and $\lambda \in \mathrm{Irr}(C)$ is linear, let

$$\mathrm{Irr}(D \mid \lambda) = \{\chi \in \mathrm{Irr}(D) : \chi|_C = \chi(1)\lambda\}.$$

Here $\mathrm{Irr}(D)$ denotes the irreducible complex characters of D. More generally, for a normal subgroup R and $\theta \in \mathrm{Irr}(R)$, $\mathrm{Irr}(D \mid \theta)$ consists of the irreducible characters whose restriction to R contains θ.

Theorem 1.2. *Let D be a finite group, let $C \leq Z(D)$ be cyclic, and let $\lambda \in \mathrm{Irr}(C)$ be faithful. Suppose that $\Delta \leq \mathrm{Aut}(D) \times \Omega$ preserves C and λ under the paired action, and acts transitively on $\mathrm{Irr}(D \mid \lambda)$. Then D is soluble.*

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Preserving λ is a condition on each pair (a, σ); neither component is required to preserve it separately. Navarro and Rizo proved the corresponding solubility result for an automorphism action, in a more general normal-subgroup setting [8, Theorem A]. Allowing Galois operations requires an additional argument, because they need not preserve irreducible Brauer characters for an arbitrary finite group.

Theorem 1.2 is proved simultaneously with its 2-Brauer analogue for groups whose soluble-radical quotient is a direct product of copies of $\mathrm{GL}_3(2)$. Canonical lifts give Galois stability of Brauer characters in this class. A single minimal counterexample, chosen across both character types, produces an irreducible affine action. Its classification forces characteristic two and a radical quotient $\mathrm{GL}_3(2)^k$. This is impossible in the Brauer case, where the affine section has odd order. In the ordinary case, restriction to odd-order elements and quotienting by a nontrivial central 2-subgroup produce a smaller Brauer counterexample.

The affine classification and Galois stability are established first, before the shared minimal-counterexample argument. The last section passes from characters to twisted group algebras. In positive characteristic, a normal complement and a lift over Witt vectors reduce the assertion to characteristic zero. The proof uses the classification of finite simple groups through the cited prime-power-index and Schur-multiplier results.

Conventions and standard ingredients. All groups in the character-theoretic arguments are finite. A group is *perfect* if it equals its commutator subgroup. For a finite group D, the notation $D^{(\infty)}$ means the final, stable term of its derived series; it is perfect, and $D/D^{(\infty)}$ is soluble. We use $F(D)$ for the Fitting subgroup, the largest nilpotent normal subgroup. A *component* is a subnormal perfect subgroup whose quotient by its center is nonabelian simple. The layer $E(D)$ is the product of the components. The generalized Fitting subgroup is $F^*(D) = F(D)E(D)$, and its centralizer is contained in it [6]. For a prime p, the symbols $O_p(D)$ and $O_{p'}(D)$ denote, respectively, the largest normal p-subgroup and the largest normal subgroup of order prime to p.

For a prime p, a p-regular element is an element whose order is prime to p, and $\mathrm{IBr}_p(D)$ denotes the irreducible p-Brauer characters. Fix compatible identifications of roots of unity of order prime to p with complex roots of unity. Brauer characters are then complex-valued functions on the p-regular elements. For an ordinary character χ, its restriction to these elements is denoted by χ^0. Pointwise application of $\sigma \in \Omega$ to a Brauer character is always defined as a function; its irreducibility will be proved whenever it is needed. The notation $\mathrm{IBr}_p(D \mid \theta)$ has the same restriction meaning as in the ordinary case. For either kind of character of a normal subgroup, its inertia group is $I_D(\theta) = \{g \in D : \theta^g = \theta\}$.

We use ordinary and modular Clifford theory and Gallagher's correspondence in their usual forms [5, Chapters 6 and 11][11, Theorems 4.6–4.8]. Thus restriction to a normal subgroup is homogeneous up to conjugation, induction from the inertia group gives the Clifford correspondence, and an extension of an invariant irreducible character identifies the characters above it with the irreducible characters of the quotient. These assertions will be applied only after the relevant invariance and extension conditions have been checked.

For a finite perfect group Q, its universal central extension $U \longrightarrow Q$ is finite and has central kernel the Schur multiplier of Q. The group U is perfect and has trivial Schur multiplier; this property is called *superperfectness*. Consequently, for the trivial actions on the coefficient groups,

$$H^2(U, \mathbb{C}^\times) = 0, \qquad H^2(U, \overline{\mathbb{F}}_2^\times) = 0.$$

An invariant irreducible representation across a quotient U therefore extends, and the extension is unique because U has no nontrivial linear characters. Automorphisms of Q lift uniquely to U; these facts give the equivariance in the descent arguments below [2].

We work in the full group Ω, rather than selecting lifts between finite cyclotomic Galois groups. Every action on a particular finite character set has finite image. Inner automorphisms can always be adjoined to a paired acting group: they fix ordinary and Brauer characters of the whole group and fix every central character.

2. An affine classification

We first isolate the finite permutation-group argument used in both character-theoretic reductions. For a finite-dimensional vector space V over $\mathbb{F}_p$, write $\mathrm{AGL}(V)$ for its affine group and $\ell : \mathrm{AGL}(V) \to \mathrm{GL}(V)$ for linear projection. We identify V with the subgroup of translations. The linear part of an affine group E is called irreducible when V is an irreducible $\mathbb{F}_p\ell(E)$-module.

Lemma 2.1. *Let $E \leq \mathrm{AGL}(V)$ be transitive, with irreducible linear part, where $V \neq 0$. Suppose that $1 \neq J \lhd E$ is perfect and that every point stabilizer in J is soluble. Then $\mathrm{Rad}(J) = 1$, and linear projection is injective on J.*

Proof. The image of $O_p(J)$ under linear projection is a normal p-subgroup of $\ell(E)$. A p-group acting in characteristic p has nonzero fixed vectors. Its fixed space is $\ell(E)$-invariant, so irreducibility makes this image trivial. Thus $O_p(J)$ consists of translations.

The translation subgroup $J \cap V$ is an $\ell(E)$-invariant subspace, and hence equals 0 or V. In the latter case $J = VJ_0$, so $J/V \cong J_0$ is soluble. It is also perfect, as a quotient of J, and therefore trivial. This would make $J = V$ abelian and perfect, contrary to $J \neq 1$. Consequently $J \cap V = 1$ and $O_p(J) = 1$.

The group $O_{p'}(J)$ fixes an affine point: average the orbit of any point, using that its order is invertible in $\mathbb{F}_p$. Its fixed-point set is E-invariant because $O_{p'}(J)$ is normal in E. Transitivity makes this set all of V, and faithfulness gives $O_{p'}(J) = 1$. If the soluble radical of J were nontrivial, its Fitting subgroup would contain a nontrivial subgroup $O_r(J)$ for some prime r. This contradicts $O_p(J) = O_{p'}(J) = 1$. □

Theorem 2.2. *Under the hypotheses of Lemma 2.1, suppose in addition that J is a direct product of nonabelian simple groups. Then there is an integer $k \geq 1$ such that*

$$p = 2, \qquad J \cong \mathrm{GL}_3(2)^k.$$

Proof. Put $n = \dim_{\mathbb{F}_p} V$ and write $J = S_1 \times \cdots \times S_k$, where each S_i is nonabelian simple. Since J is normal in the transitive group E, its orbits have equal lengths dividing p^n. Each factor S_i is normal in J, so its orbits within a J-orbit also have equal lengths dividing that p-power. The stabilizer of any point in S_i is soluble, and is proper because S_i is not soluble.

The classification of simple groups with a soluble subgroup of prime-power index gives the following possibilities for a simple factor and its orbit length:

$$(A_5, 5), \quad (\mathrm{PSL}_3(2), 7), \quad (\mathrm{PSL}_3(3), 13), \quad (\mathrm{PSL}_2(q), q+1).$$

In the last case $q + 1$ is a prime power. The actions are the natural point or hyperplane actions in the first three cases, and the projective-line action in the last case; see [1, Lemma 6 and Corollary 7]. In particular, for a fixed simple group and a fixed prime p, all the orbits have the same length.

The fixed points of an affine transformation form either the empty set or an affine subspace. A nonempty fixed-point set therefore has p-power cardinality. This elementary observation excludes three entries at once. A 3-cycle in the degree-five action of A_5 fixes two points. An involution in either degree-seven action of $\mathrm{PSL}_3(2)$ fixes three points. In either degree-thirteen action of $\mathrm{PSL}_3(3)$, the involution represented by $\mathrm{diag}(-1, -1, 1)$ fixes five points: four on its two-dimensional eigenspace and one on its complementary eigenline. On V the corresponding counts are, respectively, $2 \cdot 5^a$, $3 \cdot 7^a$ and $5 \cdot 13^a$ for some $a \geq 0$, none of which is a power of the relevant characteristic.

It remains to consider $\mathrm{PSL}_2(q)$. If q is even, a nonidentity split-torus element fixes exactly two projective points. Since the prime dividing $q + 1$ is odd, its total fixed-point count on V is again impossible. Hence q is odd; the same classification gives q prime and

$$p = 2, \qquad q + 1 = 2^m.$$

Nonabelian simplicity excludes $q = 3$, so $m \geq 3$.

Choose an element x of order q in S_i. It fixes one point in each projective-line orbit, and therefore 2^{n-m} points on V. Translating one fixed point to the origin shows that

$$\operatorname{rank}(\ell(x) - 1) = m. \tag{2.1}$$

The multiplicative order of 2 modulo $q = 2^m - 1$ is m: no smaller positive exponent can give a multiple of q. Thus every nontrivial irreducible $\mathbb{F}_2\langle x\rangle$-module has dimension m. Equation (2.1) says that the nontrivial eigencharacters of x constitute a single Frobenius orbit of length m, each occurring once. The normalizer of $\langle x\rangle$ in S_i induces the squares in $\mathbb{F}_q^\times$, a subgroup of order $(q-1)/2$, on these eigencharacters. This subgroup acts freely on the nontrivial characters and must preserve that one orbit. Hence

$$\frac{q-1}{2} \leq m.$$

Together with $q = 2^m - 1$ and $m \geq 3$, this forces $m = 3$ and $q = 7$. Thus every S_i is isomorphic to $\mathrm{GL}_3(2)$, as required. □

3. Galois stability in characteristic two

Put $S = \mathrm{GL}_3(2)$, and consider the class

$$\mathcal{C} = \{G \text{ finite} : G/\operatorname{Rad}(G) \cong S^k \text{ for some } k \geq 0\}.$$

Here $\operatorname{Rad}(G)$ is the soluble radical. In particular, $\mathcal{C}$ contains all soluble groups. We shall use this class exactly as defined; knowing only the composition factors of a group would not suffice.

We need three facts about S. Every proper subgroup of S is soluble; its Schur multiplier has order two; and its irreducible 2-Brauer characters have degrees $1, 3, 3, 8$. Each of these four Brauer characters is the restriction of an ordinary irreducible character of the same degree. Since the ordinary degrees are $1, 3, 3, 6, 7, 8$, these four lifts form a Galois-stable set. These facts are recorded in the character-table and subgroup data for S [3]. Consequently the full cyclotomic Galois group permutes $\mathrm{IBr}_2(S)$ and, by taking tensor products, also $\mathrm{IBr}_2(S^k)$.

We also use canonical lifts for soluble groups. If R is soluble, restriction to odd-order elements gives a bijection

$$B_{2'}(R) \longrightarrow \mathrm{IBr}_2(R).$$

The set on the left is invariant under group automorphisms and under all cyclotomic Galois automorphisms [11, Theorems 1.25 and 4.30]. Thus an invariant Brauer character has an invariant canonical ordinary lift, and taking canonical lifts commutes with Galois conjugation.

Lemma 3.1. *The class $\mathcal{C}$ is closed under subgroups, quotients, and finite central extensions.*

Proof. First let $H \leq S^k$. Project H onto those coordinates on which its projection is all of S. The image is a subdirect product of copies of a nonabelian simple group, hence a product of diagonal copies of S. This standard consequence of Goursat's lemma follows by adding one coordinate at a time: the new coordinate either supplies an independent factor or is identified with an existing simple quotient. The kernel embeds in a product of proper subgroups of S, so it is soluble. Since the image has trivial soluble radical, the kernel is precisely $\operatorname{Rad}(H)$.

For a subgroup of an arbitrary $G \in \mathcal{C}$, apply this observation to its image in $G/\operatorname{Rad}(G)$. The additional kernel lies in $\operatorname{Rad}(G)$ and is soluble. Quotient closure follows because every quotient of S^k is a product of some of its simple factors. Finally, in a central extension of G, the inverse image of $\operatorname{Rad}(G)$ is soluble and its quotient is S^k; it is therefore the soluble radical of the extension. □

Brauer characters will be regarded as complex-valued functions on the odd-order elements, using a fixed identification of odd-order roots of unity with their modular counterparts. For an arbitrary group, pointwise Galois conjugation need not preserve irreducible Brauer characters. The next result establishes the precise stability needed here.

Theorem 3.2. *Let $G \in \mathcal{C}$. For every $\varphi \in \mathrm{IBr}_2(G)$ and every cyclotomic Galois automorphism σ, the function φ^σ belongs to $\mathrm{IBr}_2(G)$.*

Proof. We argue by induction on $|G|$. For soluble groups the assertion follows from the canonical lifting bijection. Set $R = \mathrm{Rad}(G)$, choose a constituent θ of φ_R, and let $T = I_G(\theta)$.

If $T < G$, modular Clifford theory writes $\varphi = \psi^G$ for a unique $\psi \in \mathrm{IBr}_2(T \mid \theta)$. The closure lemma and induction give $\psi^\sigma \in \mathrm{IBr}_2(T)$. Moreover

$$I_G(\theta^\sigma) = I_G(\theta) = T,$$

because Galois conjugation commutes with group conjugation and is invertible. The restriction of ψ^σ to R is a multiple of θ^σ. Clifford theory therefore makes $(\psi^\sigma)^G$ irreducible, and induction of characters commutes with Galois conjugation. Hence $\varphi^\sigma = (\psi^\sigma)^G$ is irreducible.

Suppose now that θ is G-invariant, and let $\chi \in B_{2'}(R)$ be its canonical lift. This ordinary character is also G-invariant. Let $U \to G/R$ be the universal central extension, with kernel M, and form the fibre product

$$\widehat{G} = G \times_{G/R} U.$$

The quotient $\widehat{G}/R$ is U. Since U is superperfect, the obstruction in $H^2(U, \mathbb{C}^\times)$ vanishes and χ extends to an ordinary irreducible character $\widehat{\chi}$ of $\widehat{G}$. The extension is unique because U has no nontrivial linear characters. Furthermore, $M = \ker(\widehat{G} \to G)$ is a central 2-group: the multiplier of S^k is the product of the multipliers of its factors.

Choose an invariant lattice affording $\widehat{\chi}$ in a sufficiently large 2-modular system, and reduce it modulo its maximal ideal. Its restriction to R has Brauer character $\chi^0 = \theta$, so that restriction, and therefore the entire reduced module, is irreducible. The central group M acts on the ordinary representation by scalars of 2-power order. These scalars reduce to one. The reduced module thus descends to an irreducible Brauer character τ of G extending θ.

Apply the same reasoning to $\widehat{\chi}^\sigma$. Its restriction χ^σ is the canonical lift of θ^σ, so its reduction again descends to an irreducible character of G. This character is τ^σ. Indeed, an odd-order element $g \in G$ has a unique odd-order lift $\widetilde{g}$ in $\widehat{G}$: the inverse image of $\langle g \rangle$ is abelian, and its odd Hall subgroup maps isomorphically onto $\langle g \rangle$. Thus

$$\tau(g) = \widehat{\chi}(\widetilde{g}), \qquad \tau^\sigma(g) = \widehat{\chi}^\sigma(\widetilde{g}).$$

Finally, the modular Gallagher correspondence gives

$$\mathrm{IBr}_2(G \mid \theta) = \{\tau\beta : \beta \in \mathrm{IBr}_2(G/R)\},$$

with quotient characters inflated to G. The Galois stability of $\mathrm{IBr}_2(S^k)$ now shows that every $(\tau\beta)^\sigma = \tau^\sigma\beta^\sigma$ is irreducible. □

The Galois operations in this proof are performed on ordinary characters before reduction to characteristic two.

Restriction transfers transitivity.

Lemma 3.3. *Let $D/\mathrm{Rad}(D)$ be a direct product of copies of $\mathrm{GL}_3(2)$. Let $C \le Z(D)$ be cyclic, let $\lambda \in \mathrm{Irr}(C)$ be faithful, and suppose that a paired group Δ preserves C and λ and is transitive on $\mathrm{Irr}(D \mid \lambda)$. Then Δ is transitive on $\mathrm{IBr}_2(D \mid \lambda|_{C_{2'}})$, where $C_{2'}$ is the odd-order Hall subgroup of C.*

Proof. Write $C = C_2 \times C_{2'}$, where C_2 is the Sylow 2-subgroup, and write $\lambda = \lambda_2 \times \lambda_{2'}$. Theorem 3.2 ensures that Δ permutes

$$X = \mathrm{Irr}(D \mid \lambda), \qquad Y = \mathrm{IBr}_2(D \mid \lambda_{2'}).$$

Restriction to odd-order elements defines a linear map

$$\mathbb{C}X \longrightarrow \mathbb{C}Y.$$

We verify surjectivity explicitly. The space $\mathbb{C}X$ consists of the class functions f on D satisfying $f(cg) = \lambda(c)f(g)$ for $c \in C$. The corresponding description of $\mathbb{C}Y$ uses class

functions on the odd-order elements and the character $\lambda_{2'}$ of $C_{2'}$. For such a target function f, use the commuting primary decomposition $g = g_2 g_{2'}$ and set

$$\widetilde{f}(g) = \begin{cases} \lambda_2(g_2) f(g_{2'}), & g_2 \in C_2, \\ 0, & g_2 \notin C_2. \end{cases}$$

This is conjugation-invariant. Since elements of C are central, multiplication by $c = c_2 c_{2'}$ replaces the primary factors by $c_2 g_2$ and $c_{2'} g_{2'}$. Thus $\widetilde{f}(cg) = \lambda(c)\widetilde{f}(g)$. Its restriction to the odd-order elements is f, proving surjectivity.

The actions on the two vector spaces are the *complex-linear* extensions of their permutation actions on irreducible characters. Restriction intertwines these actions. Indeed, the decomposition numbers in

$$\chi^0 = \sum_{\varphi \in Y} d_{\chi\varphi}\varphi$$

are integers. Applying σ to character values preserves these coefficients and gives the corresponding equality for χ^σ; group automorphisms commute with restriction as well. Dualizing this surjection embeds the space of invariant functionals on $\mathbb{C}Y$ into that on $\mathbb{C}X$. For a permutation basis, this dimension is the number of orbits. Since X is transitive and Y is nonempty, Y has exactly one orbit. □

4. One argument for ordinary and Brauer characters

We prove Theorem 1.2 together with the following modular analogue.

Theorem 4.1. *Let $G \in \mathcal{C}$, let $C \le Z(G)$ be cyclic of odd order, and let $\lambda \in \mathrm{Irr}(C)$ be faithful. Suppose a group of paired automorphism–Galois operations preserves C, fixes λ, and is transitive on $\mathrm{IBr}_2(G \mid \lambda)$. Then G is soluble.*

Lemma 4.2. *Work either with ordinary characters of finite groups, or with irreducible 2-Brauer characters of groups in $\mathcal{C}$. Let $R \triangleleft N$ and let θ be an N-invariant irreducible character of R of the chosen kind. If N/R is perfect, there exist a finite perfect group D_0, a cyclic central subgroup C_0, and a faithful $\lambda_0 \in \mathrm{Irr}(C_0)$, such that $D_0/C_0 \cong N/R$ and the characters of N above θ correspond bijectively to the characters of D_0 above λ_0. The correspondence is equivariant for paired operations preserving R and fixing θ. In the Brauer case, $D_0 \in \mathcal{C}$ and C_0 has odd order.*

Proof. Let $U \twoheadrightarrow N/R$ be the universal central extension, with kernel M, and form the pullback $\widehat{N} = N \times_{N/R} U$. Thus $\widehat{N}/R \cong U$, while the kernel of $\widehat{N} \twoheadrightarrow N$ is the central subgroup M. The group U is finite and superperfect. Hence the character-extension obstruction vanishes, in $H^2(U, \mathbb{C}^\times)$ or $H^2(U, \overline{\mathbb{F}}_2^\times)$ respectively. The resulting extension $\widehat{\theta}$ is unique, because any two extensions differ by a linear character of the perfect group U.

Let κ be the scalar character of M in this extension. In the Brauer case it has odd order and is viewed as a complex linear character trivial on the Sylow 2-subgroup of M. The ordinary or modular Gallagher correspondence gives the characters above θ in $\widehat{N}$ by tensoring its extension with irreducible representations of U. Such a tensor product descends to N precisely when its scalar character on M is trivial. We may therefore take

$$D_0 = U/\ker\kappa, \qquad C_0 = M/\ker\kappa,$$

with λ_0 induced by κ^{-1}. These have the required properties; in the Brauer case, class membership follows from Lemma 3.1.

An automorphism of N preserving R induces one of N/R, which lifts uniquely to U and gives a natural automorphism of the pullback. In the Brauer case $\widehat{N} \in \mathcal{C}$, so Theorem 3.2 ensures that paired transforms of $\widehat{\theta}$ remain irreducible. A pair fixing θ therefore fixes its unique extension, and hence fixes κ and preserves its kernel. The tensor-product correspondence and its descent are equivariant. □

Proof of Theorems 1.2 and 4.1. Suppose that one of the theorems fails. Among all nonsoluble configurations of either kind choose one minimizing

$$(|G : C|, |G|)$$

lexicographically. Throughout this proof, "characters" means characters of its chosen kind. Adjoin inner automorphisms to the paired group Δ, and let $\mathcal{A}$ be its projection to $\mathrm{Aut}(G)$. Invariance below means invariance under $\mathcal{A}$. In the Brauer case all subgroups, quotients, and central extensions used below belong to $\mathcal{C}$, and their characters are Galois-stable, by Lemma 3.1 and Theorem 3.2.

Minimality and restriction. For every invariant normal subgroup L, the paired action is transitive on the characters of L above $\lambda|_{C\cap L}$. Indeed, each such representation ρ extends to CL by $\widetilde{\rho}(cl) = \lambda(c)\rho(l)$, using the corresponding odd-order modular roots of unity in the Brauer case. This extension is covered by a character of G. Transitivity on the covering characters and Clifford theory connect any two constituents on L, using an inner conjugation to align the constituents.

Apply this observation to the last derived subgroup $G^{(\infty)}$. Its index over $C \cap G^{(\infty)}$ is at most $|G : C|$, and its order is smaller if it is proper. Minimality would make it, and hence G, soluble. Thus G is perfect. In the Brauer case every irreducible character kills $O_2(G)$. Since C is odd, quotienting by a nontrivial $O_2(G)$ strictly decreases the central index and preserves nonsolubility. Therefore $O_2(G) = 1$, and in particular $Z(G)$ is odd.

We claim that $C = Z(G)$. Choose a linear character ν of $Z(G)$ above λ. Its paired stabilizer is transitive on the characters of G above ν. Factoring by $\ker\nu$ gives a faithful cyclic central character and central index $|G : Z(G)|$. If $Z(G) > C$, minimality makes this quotient soluble, and its central kernel then makes G soluble. This quotient action is well defined: Galois conjugation preserves the kernel of a linear character, so the paired stabilizer preserves $\ker\nu$.

Every proper invariant normal subgroup L is soluble. If $CL < G$, restriction and minimality applied to CL give this assertion. The alternative $CL = G$ would make the perfect quotient G/L abelian and force $L = G$. Consequently $G/\operatorname{Rad}(G)$ is characteristically simple: the inverse image of any proper characteristic subgroup is soluble, while this quotient has trivial soluble radical. Thus $G/\operatorname{Rad}(G)$ is a direct product of isomorphic nonabelian simple groups.

Soluble inertia groups. Let $C < R < G$ be invariant and normal, and let θ be a character of R above λ. Put $T = I_G(\theta)$ and $N = RT^{(\infty)}$. Clifford induction makes the paired stabilizer Δ_θ transitive on the characters of T above θ: map their induced characters in G, correct by an inner conjugation to fix θ, and use uniqueness of the Clifford correspondent. This stabilizer preserves N and contains the inner automorphisms of T. Restriction and Clifford theory therefore give transitivity on the characters of N above θ.

The quotient N/R is perfect. Lemma 4.2 produces a configuration of the same kind whose central index is

$$|D_0 : C_0| = |N : R| \le |G : R| < |G : C|.$$

Minimality makes D_0 soluble, and its perfectness gives $D_0 = 1$. Thus $T^{(\infty)} \le R$. Since R is soluble, so is T. In particular, no such R has a unique character above λ, because its inertia group would be G.

Eliminating the layer. In the ordinary case we first show

$$\text{every prime dividing } |G| \text{ divides } |C|. \tag{4.1}$$

Otherwise, for a prime $q \mid |G|$ with $q \nmid |C|$, choose a Sylow q-subgroup P. The subgroup $CP = C\times P$ has linear character $\rho = \lambda\times 1_P$. Pairs normalizing P fix ρ, and are still transitive on $\mathrm{Irr}(G \mid \lambda)$: Sylow conjugacy adjusts any pair by an inner automorphism without changing its action on characters of G. If there are r characters above λ, all have the same degree d

and the same positive multiplicity b in ρ^G. The degree and degree-square identities give

$$|G : CP| = brd, \qquad |G : C| = rd^2.$$

The first index is prime to q, whereas the second is divisible by q, a contradiction.

The layer $E(G)$ is characteristic and perfect, so it is either trivial or all of G. Suppose that $G = E(G)$. Then G is a central product of quasisimple groups, its radical is its center, and all the components have the same simple central quotient. In the ordinary case this simple group has a prime divisor absent from its Schur multiplier [8, Theorem 3.4]. The centers of the components are quotients of that multiplier and generate $Z(G)$. This prime therefore divides $|G|$ but not $|C|$, contradicting (4.1). In the Brauer case the simple quotient is $S = \mathrm{GL}_3(2)$, whose multiplier has order two. Hence $Z(G)$ is a 2-group. Since $O_2(G) = 1$, we obtain $G \cong S^k$ and $C = 1$. Its trivial Brauer character is fixed by every pair, whereas nontrivial irreducible Brauer characters exist. This again contradicts transitivity. Thus $E(G) = 1$ in both cases.

An abelian normal section. The generalized Fitting subgroup is now $F = F(G)$, so $C_G(F) \leq F$. It follows that $C < F < G$: equality $F = C$ would give $G \leq F$, and $F = G$ would make G soluble. In the Brauer case F is odd because $O_2(G) = 1$.

In each Sylow subgroup of F choose a characteristic Thompson critical subgroup. Their product T is characteristic, has class at most two, and satisfies $C_F(T) = Z(T)$ [4]. If $Z(T) \leq C$, set $R = CT$. Then $Z(R) = C$ and $C < R < G$. Indeed, $T \leq C$ would give $F = C_F(T) = Z(T) \leq C$. For $\chi \in \mathrm{Irr}(R \mid \lambda)$ and $x \notin C$, choose $y \in R$ with $[x, y] \neq 1$. This commutator lies in C, so

$$\chi(x) = \chi(x^y) = \lambda([x, y])\chi(x).$$

Faithfulness gives $\lambda([x, y]) \neq 1$, so $\chi(x) = 0$. Orthogonality gives $\chi(1)^2 = |R : C|$, and the degree-square sum above λ then makes χ unique. In the Brauer case R is odd, so ordinary and Brauer characters coincide. In either case uniqueness contradicts the inertia-group conclusion.

Therefore $CZ(T) > C$. Choose A/C minimal nontrivial invariant in $CZ(T)/C$. Then A is abelian, $C < A < G$, and A/C is an elementary abelian q-group irreducible under $\mathcal{A}$. In the Brauer case q is odd.

The affine contradiction. The set $\mathrm{Irr}(A \mid \lambda)$ is an affine space with translation space $\mathrm{Hom}(A/C, \mu_q)$, where μ_q denotes the complex q-th roots of unity. In the Brauer case these are also the Brauer characters of A above λ. Restriction gives a transitive paired action with finite image E. Its linear parts are dual automorphisms of A/C multiplied by Galois scalars, so its linear part is irreducible.

Let J be the image of the inner automorphisms of G in E. It is perfect and normal. It is nontrivial because the characters above λ separate A: their ratios separate A/C, and λ is faithful on C. If $J = 1$, every commutator $[a, g]$ would lie in all their kernels, giving $A \leq Z(G) = C$. The point stabilizers of J are images of the soluble inertia groups $I_G(\nu)$, with $\nu \in \mathrm{Irr}(A \mid \lambda)$. Lemma 2.1 therefore gives $\mathrm{Rad}(J) = 1$. The kernel of $G \twoheadrightarrow J$ is a proper invariant normal subgroup, hence soluble; since J has trivial radical, that kernel is exactly $\mathrm{Rad}(G)$. Thus J is a direct product of nonabelian simple groups, and Theorem 2.2 gives

$$q = 2, \qquad G/\mathrm{Rad}(G) \cong J \cong \mathrm{GL}_3(2)^k \quad (k \geq 1).$$

This immediately contradicts the oddness of q in the Brauer case.

It remains to consider the ordinary case. We have now proved $G \in \mathcal{C}$, so Lemma 3.3 transfers the paired transitivity to $\mathrm{IBr}_2(G \mid \lambda|_{C_{2'}})$. Since $|G|$ is even, (4.1) shows that the Sylow 2-subgroup C_2 of C is nontrivial. Every irreducible 2-Brauer character kills C_2, so the paired action descends to the modular configuration

$$G/C_2, \qquad C/C_2, \qquad \overline{\lambda},$$

where $\overline{\lambda}$ is the faithful character induced by $\lambda|_{C_{2'}}$. The action descends because C_2 is characteristic in the preserved subgroup C. The quotient is still nonsoluble and belongs to $\mathcal{C}$.

Its parameters satisfy

$$|G/C_2 : C/C_2| = |G : C|, \qquad |G/C_2| < |G|.$$

This contradicts our common minimal choice and proves both theorems. □

5. Twisted group algebras over arbitrary fields

5.1. **Characteristic zero.** Assume that $\operatorname{char} K = 0$ and that $K^\alpha G$ is simple. Let k be the subfield generated over $\mathbb{Q}$ by the finitely many coefficients of α. The algebra $k^\alpha G$ is simple: extending a nonzero proper ideal to K would give a nonzero proper ideal of $K^\alpha G$, by faithful flatness of field extensions. The field k is finitely generated over $\mathbb{Q}$, so we may embed an algebraic closure $\overline{k}$ into $\mathbb{C}$.

Put $n = |G|$ and $b(g) = \prod_{x \in G} \alpha(g, x)$. Multiplying the cocycle identity over $x \in G$, and using the permutation $x \mapsto hx$, gives

$$\alpha(g, h)^n b(gh) = b(g)b(h).$$

Choose $t(g) \in \overline{k}^\times$ with $t(g)^n = b(g)$ and $t(1) = 1$. Then

$$\beta(g, h) = \alpha(g, h)\frac{t(gh)}{t(g)t(h)}$$

is a normalized cocycle with values in the group μ_n of n-th roots of unity. Define the finite group

$$H = \mu_n \times_\beta G, \qquad (z, g)(w, h) = (zw\beta(g, h), gh).$$

Its subgroup μ_n is central and $H/\mu_n \cong G$. Inside the units of $\overline{k}^\alpha G$, the element (z, g) is represented by $zu_g/t(g)$. Hence, with $\lambda_0(z) = z$, the simple $\overline{k}^\alpha G$-modules correspond to $\operatorname{Irr}(H \mid \lambda_0)$.

The algebra $k^\alpha G$ is finite-dimensional and simple. Its center is a finite separable extension of k, and its simple components after extension to $\overline{k}$ form one orbit under $\operatorname{Gal}(\overline{k}/k)$. To describe the resulting action on characters correctly, let $\sigma \in \operatorname{Gal}(\overline{k}/k)$ and set $\ell_\sigma(g) = \sigma(t(g))/t(g) \in \mu_n$. Applying σ to the embedded group of units gives the automorphism

$$f_\sigma(z, g) = (\sigma(z)\ell_\sigma(g)^{-1}, g).$$

Thus the transitive action on $\operatorname{Irr}(H \mid \lambda_0)$ is

$$\chi \longmapsto \chi^\sigma \circ f_\sigma^{-1}.$$

This formula follows by transporting a representation along the semilinear action on $\overline{k}^\alpha G$.

Because these automorphisms arise from an actual Galois action on the algebra, the map

$$\sigma \longmapsto (f_\sigma, \sigma|_{\mathbb{Q}^{\mathrm{ab}}})$$

is a homomorphism into $\operatorname{Aut}(H) \times \Omega$. Its image preserves μ_n and fixes λ_0 as a pair: on μ_n, the first component acts by σ, which is cancelled by the Galois operation on character values. The transitivity just established therefore satisfies exactly the hypotheses of Theorem 1.2. That theorem makes H soluble, and its quotient G is soluble.

5.2. **Positive characteristic.** We first record a lifting fact that makes the passage to characteristic zero explicit.

Lemma 5.1. *Let R be a discrete valuation ring with uniformizer π and fraction field F. Let A be an R-algebra that is finite free as an R-module. If $A/\pi A$ is simple, then $F \otimes_R A$ is simple.*

Proof. Suppose that I is a nonzero proper two-sided ideal of $F \otimes_R A$, and put $J = I \cap A$. Clearing denominators shows that $FJ = I$. Thus J has positive rank, and A/J has positive rank because I is proper. The submodule J is saturated: if $\pi a \in J$ for $a \in A$, then $a \in I$ and hence $a \in J$. In particular, $J \cap \pi A = \pi J$ and A/J is torsion-free. Finitely generated

torsion-free modules over a discrete valuation ring are free. It follows that $J/\pi J$ is a nonzero proper two-sided ideal of $A/\pi A$, a contradiction. □

Now assume $\operatorname{char} K = p > 0$. Passman's normal-complement and Hall-subalgebra results [9, Lemmas 2.6(ii) and 2.7] give a normal subgroup $N \triangleleft G$ such that $p \nmid |N|$, G/N is a p-group, and $K^\alpha N$ is simple. It is enough to prove that N is soluble.

Since $|N|$ is invertible in K, the algebra $K^\alpha N$ is separable over K. One separability element is

$$\frac{1}{|N|} \sum_{g \in N} u_g \otimes u_g^{-1}.$$

Indeed, its product is 1, and multiplication by a basis unit on either side agrees after reindexing the sum. The center of $K^\alpha N$ is therefore a finite separable field extension of K.

Let k be the perfect closure of K. A finite separable extension is linearly disjoint from a purely inseparable one; thus the center remains a field after extension to k. Since a central simple algebra stays central simple after extending its center, $k^\alpha N$ remains simple.

The Witt ring $W(k)$ is a complete discrete valuation ring of characteristic zero with residue field k and uniformizer p [10, Chapter II]. Its multiplicative Teichmüller map lifts α coefficientwise to a cocycle $\widetilde{\alpha}$ with unit coefficients in $W(k)$. The algebra $W(k)^{\widetilde{\alpha}} N$ is finite free, and its reduction modulo p is $k^\alpha N$. Lemma 5.1 makes the twisted group algebra of N over the fraction field of $W(k)$ simple. The characteristic-zero argument makes N soluble. Finally, G/N is a p-group, so G is soluble. This proves Theorem 1.1.

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